

Device Agent — Setup & Operations Guide

Cross-platform (Windows · macOS · Linux) · Phase 1

Audience IT admins installing the Miles Agent on a managed laptop and verifying it from the portal.

1. Prerequisites

Done once by the IT team — verify before rolling out:

Component	Status / Action
Supabase migration 20260527000000_device_management	Run once in Supabase SQL Editor. On success, the portal returns: "Could not find function generate_agent_token"
Edge function agent-api	Already deployed at https://dimbgprindvmzoylzyud.supabase.co/functions/v1/agent-api
Portal	Live at https://it-asset-hub-a7rf.onrender.com
Admin role	User generating keys must have super_admin role

2. Generate an Agent Key (in the portal)

- Log into the portal as **Super Admin**.
- Open the laptop asset (Assets → click the row).
- Right column → **Device Agent** card → click **Generate Agent Key**.
- A dialog shows the full key (starts with `mi l_...`) — **copy it now**.

Important The full key is shown only once. After closing the dialog only the last 4 chars remain visible. If lost, click **Regenerate Key** to issue a new one.

3. Install the agent on the laptop

Two options. Use (A) for quick testing, (B) to ship a binary to end users.

Option A · Run the Python script directly

Requires Python 3.10+ installed. Copy `laptop_agent.py` to the laptop, then:

```
# Windows (PowerShell as Administrator)
pip install requests
$env:MILES_AGENT_TOKEN = "mil_paste_real_key_here"
python laptop_agent.py info          # preview what will be sent (no network)
python laptop_agent.py register      # bind to portal
python laptop_agent.py run           # long-running sync loop

# macOS / Linux
pip3 install requests
export MILES_AGENT_TOKEN="mil_paste_real_key_here"
python3 laptop_agent.py register
```

Option B · Build a single binary with PyInstaller

PyInstaller cannot cross-compile — build on each target OS.

```
pip install pyinstaller requests
pyinstaller --onefile --noconsole --name miles-agent laptop_agent.py
# → dist/miles-agent.exe (Windows)
# → dist/miles-agent      (macOS / Linux)
```

4. Set the token & register, per OS

Windows — Command Prompt as Administrator (recommended)

```
setx MILES_AGENT_TOKEN "mil_XXXXXXXXXXXXXXXXXX" /M
:: Close THIS window, open a NEW Admin Command Prompt:
echo %MILES_AGENT_TOKEN%
miles-agent.exe register
```

Common mistake In Windows PowerShell 5.1 the `&&` operator does NOT work — use a newline or `;`. Also: typing the key alone on a line will fail with *'mil_...' is not recognized as an internal or external command* — the key is a **value**, not a command.

macOS — Terminal

```
echo 'export MILES_AGENT_TOKEN="mil_XXXXXXXXXXXXXXXXXX"' >> ~/.zshrc
echo 'export MILES_EMPLOYEE_EMAIL="user@mileseducation.com"' >> ~/.zshrc
source ~/.zshrc
./miles-agent register
```

Linux / Ubuntu

```
echo 'export MILES_AGENT_TOKEN="mil_XXXXXXXXXXXXXXXXXX"' | \
  sudo tee /etc/profile.d/miles-agent.sh
source /etc/profile.d/miles-agent.sh
sudo ./miles-agent register # sudo needed once for BIOS serial number
```

5. Schedule background sync (every 5 minutes)

Windows — Task Scheduler

```
schtasks /Create /TN "MilesAgent" /SC MINUTE /MO 5 ^  
/TR "C:\Path\To\miles-agent.exe run" /RL HIGHEST /F
```

macOS — launchd

Save as ~/Library/LaunchAgents/com.miles.agent.plist:

```
<?xml version="1.0" encoding="UTF-8"?>  
<plist version="1.0"><dict>  
  <key>Label</key><string>com.miles.agent</string>  
  <key>ProgramArguments</key>  
  <array><string>/usr/local/bin/miles-agent</string><string>run</string></array>  
  <key>RunAtLoad</key><true/>  
  <key>KeepAlive</key><true/>  
</dict></plist>  
  
launchctl load ~/Library/LaunchAgents/com.miles.agent.plist
```

Linux — systemd user unit

Save as ~/.config/systemd/user/miles-agent.service:

```
[Unit]  
Description=Miles IT Assets Agent  
After=network-online.target  
  
[Service]  
ExecStart=/usr/local/bin/miles-agent run  
Restart=always  
RestartSec=30  
  
[Install]  
WantedBy=default.target  
  
systemctl --user daemon-reload  
systemctl --user enable --now miles-agent.service
```

6. Verify it's working

From the laptop

```
miles-agent info      # prints the JSON it would send – no network  
miles-agent sync      # sends one sync now
```

From the portal

- Refresh the laptop's asset page.
- **Device Agent** card flips to **Active** with last-sync time.
- Asset fields (Serial Number, Model, RAM, Processor, OS) auto-populate.

From Supabase (debugging)

```
-- recent syncs
select * from device_sync_logs order by created_at desc limit 20;

-- managed devices last seen
select serial_number, hostname, last_sync_at, status
  from managed_devices
  order by last_sync_at desc;
```

Tip Edge function logs are visible at Supabase Dashboard → Edge Functions → **agent-api** → Logs.

7. Troubleshooting

Symptom	Cause	Fix
Portal: "Could not find function generate_token" not applied	SQL migration not applied	Run 20260527000000_device_management.sql in Supabase
PowerShell: "The token '&&' is not a valid token" or "PowerShell 5.1 does not support &&"	PowerShell 5.1 does not support &&	Run the two commands on separate lines or use ;
CMD: "'mil_...' is not recognized as an internal or external command"	Typed the key as a command	Use: setx MILES_AGENT_TOKEN "mil_..." /M
Agent: ERROR: MILES_AGENT_TOKEN is not set in the current shell session	It is not set	On Windows, reopen the terminal after setx. On Mac/Linux, s
Register returns http 401 / Invalid token	Token revoked or mistyped	Click Regenerate Key in the portal and use the new value
Linux register: empty serial number	DMI table is root-protected	Run register once with sudo
macOS wallpaper command fails	System Events automation denied	Allow when macOS prompts (System Settings → Privacy →
Card still shows "Not Installed" after register	Browser cached the old page	Hard refresh: Ctrl/Cmd + Shift + R

8. Daily operations

- **Revoke a laptop** — open the asset → Device Agent → **Revoke**. Agent stops syncing immediately.
- **Re-issue** — click **Regenerate Key**. Old key dies, install the new one on the device.
- **Force a sync** — Admin: click *Sync Now* on the card (queues a command, picked up on next poll). User: run `miles-agent sync`.
- **Rotate tokens** — recommended every 90 days.

9. Security notes

- The token is the only secret on the machine. Revoking it from the portal stops sync immediately.
- All traffic is HTTPS to the Supabase Edge Function.
- The agent never receives or stores admin credentials.
- Phase 1 supports only safe commands: `sync_now`, `collect_system_info`, `update_wallpaper`. No wipe/reset.